

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 03

Continuity of Probability

Def: A sequence, $\{E_n\}_{n \geq 1}$, of events are said to be increasing if

$$E_n \subseteq E_{n+1}.$$

Def: A sequence, $\{E_n\}_{n \geq 1}$, of events are said to be decreasing if

$$E_{n+1} \subseteq E_n.$$

Def: For an increasing sequence, $\{E_n\}_{n \geq 1}$, of events, define

$$\lim_{n \rightarrow \infty} E_n = \bigcup_{n=1}^{\infty} E_n.$$

Def: For a decreasing sequence, $\{E_n\}_{n \geq 1}$, of events, define

$$\lim_{n \rightarrow \infty} E_n = \bigcap_{n=1}^{\infty} E_n.$$

Continuity of Probability

Theorem: [Continuity from Below] Let $\{E_n\}_{n \geq 1}$ be an increasing sequence of events, then

$$P\left(\lim_{n \rightarrow \infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n).$$

Theorem: [Continuity from Above] Let $\{E_n\}_{n \geq 1}$ be a decreasing sequence of events, then

$$P\left(\lim_{n \rightarrow \infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n).$$

► Finite additivity and continuity from above implies countable additivity.

Conditional Probability

► A die is thrown twice. What is the probability that the sum is 6?

Ans: $5/36$

► Now suppose you have observed the outcome of the first throw and it is 4. Now what is the probability that the sum will be 6?

Ans: $1/6$.

Once you are given some information or you observe something, the sample space changes. Conditional probability is a probability on the changed sample space.

Def: Let H be an event with $P(H) > 0$. For any arbitrary event A , the conditional probability of A given H is defined by

$$P(A|H) = \frac{P(A \cap H)}{P(H)}.$$

$$P(A \cap B) = \begin{cases} P(A)P(B|A) & \text{if } P(A) > 0 \\ P(B)P(A|B) & \text{if } P(B) > 0 \end{cases}$$

Def: A collection of events $\{E_1, E_2 \dots\}$ is said to be mutually exclusive if $E_i \cap E_j = \phi, \forall i \neq j$. It is said to be exhaustive if $\cup_i E_i = \mathcal{S}$.

Theorem: Let $\{E_1, E_2 \dots\}$ be a collection of mutually exclusive and exhaustive events with $P(E_i) > 0, \forall i$. Then for any event E ,

$$P(E) = \sum_i P(E|E_i)P(E_i).$$

Theorem: Let $\{E_1, E_2 \dots\}$ be a collection of mutually exclusive and exhaustive events with $P(E_i) > 0, \forall i$. Let E be any event with $P(E) > 0$. Then

$$P(E_i|E) = \frac{P(E|E_i)P(E_i)}{\sum_j P(E|E_j)P(E_j)} \quad i = 1, 2, \dots$$

Example 1: There are 3 boxes. Box 1 containing 1 white, 4 black balls. Box 2 containing 2 white, 1 black ball. Box 3 containing 3 white, 3 black balls. First you throw a fair die. If the outcomes are 1, 2 or 3 then box 1 is chosen, if the outcome is 4 then box 2 is chosen and if the outcome is 5 or 6 then box 3 is chosen. Finally you draw a ball at random from the chosen box.

- Given the drawn ball is white what is the (conditional) probability that the ball is from box 1.
- Given the drawn ball is white what is the (conditional) probability that the ball is from box 2.